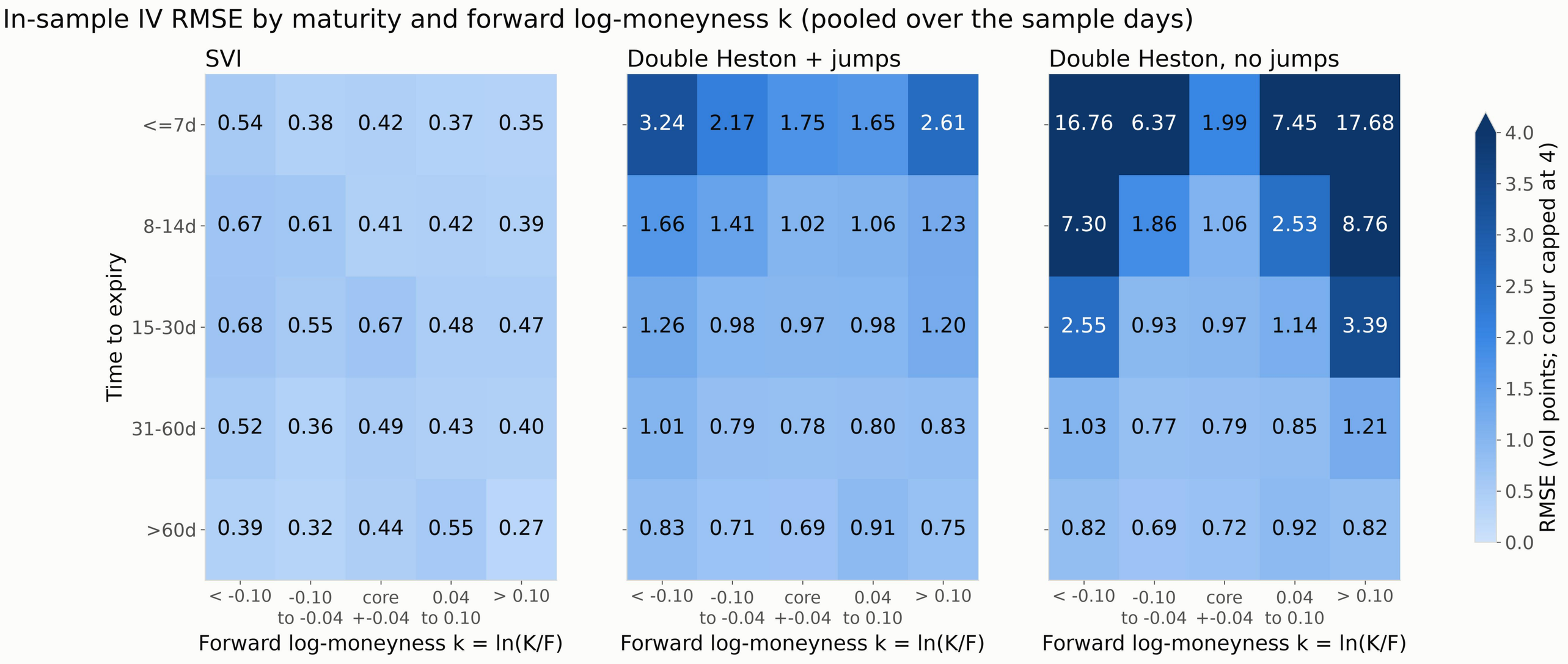

In-sample IV RMSE by maturity and forward log-moneyness k (pooled over the sample days)
SVI
Double Heston + jumps
Double Heston, no jumps
Time to expiry
<=7d
8-14d
15-30d
31-60d
>60d
0.54 0.38 0.42 0.37 0.35
0.67 0.61 0.41 0.42 0.39
0.68 0.55 0.67 0.48 0.47
0.52 0.36 0.49 0.43 0.40
0.39 0.32 0.44 0.55 0.27
3.24 2.17 1.75 1.65 2.61
1.66 1.41 1.02 1.06 1.23
1.26 0.98 0.97 0.98 1.20
1.01 0.79 0.78 0.80 0.83
0.83 0.71 0.69 0.91 0.75
16.76 6.37 1.99 7.45 17.68
7.30 1.86 1.06 2.53 8.76
2.55 0.93 0.97 1.14 3.39
1.03 0.77 0.79 0.85 1.21
0.82 0.69 0.72 0.92 0.82
< -0.10
-0.10 to -0.04
core +-0.04
0.04 to 0.10
> 0.10
Forward log-moneyness k = ln(K/F)
RMSE (vol points; colour capped at 4)
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
4.0